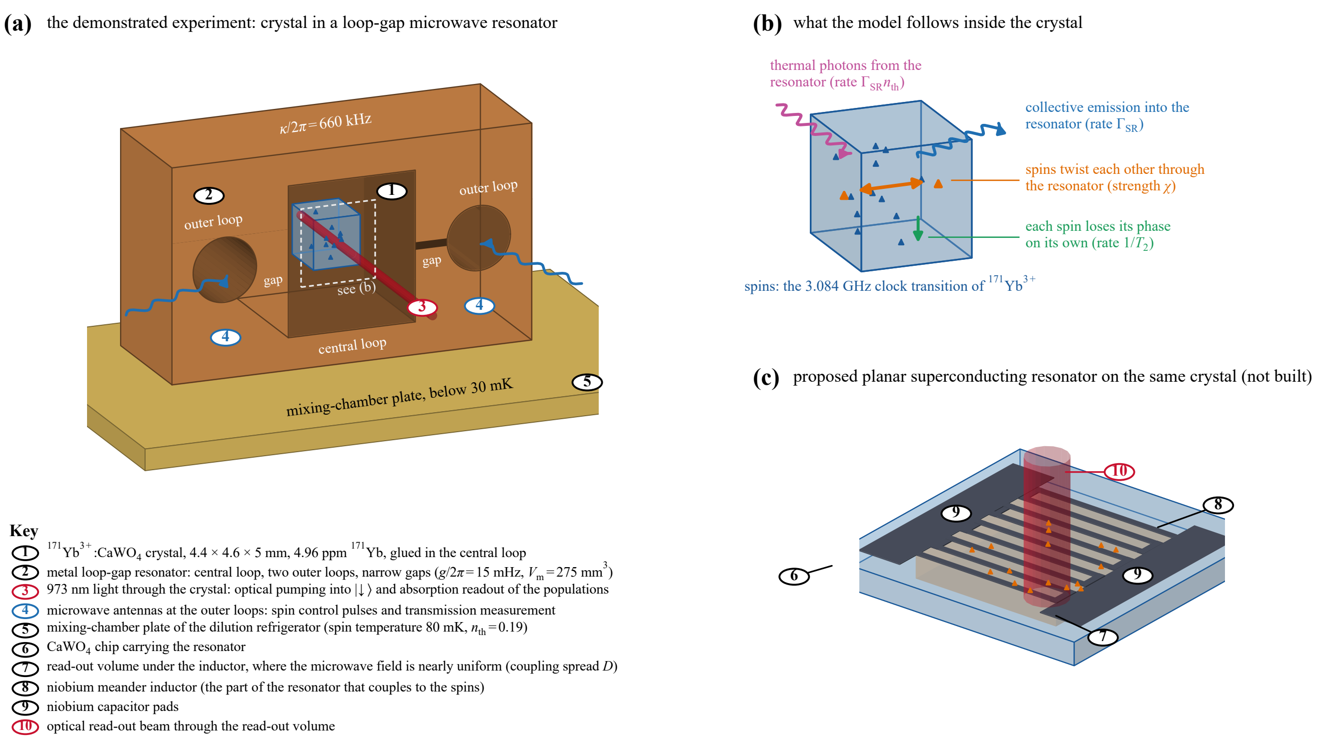
(a) the demonstrated experiment: crystal in a loop-gap microwave resonator
κ/2π = 660 kHz
outer loop
outer loop
gap
gap
see (b)
central loop
mixing-chamber plate, below 30 mK
1
2
3
4
4
5
(b) what the model follows inside the crystal
thermal photons from the resonator (rate $\Gamma_{SR} n_{th}$)
collective emission into the resonator (rate Γ_{SR})
spins twist each other through the resonator (strength χ)
each spin loses its phase on its own (rate $1/T_2$)
spins: the 3.084 GHz clock transition of $^{171}Yb^{3+}$
(c) proposed planar superconducting resonator on the same crystal (not built)
6
7
8
9
9
10
Key
1 $^{171}Yb^{3+}$:$CaWO_4$ crystal, 4.4 × 4.6 × 5 mm, 4.96 ppm ^{171}Yb, glued in the central loop
2 metal loop-gap resonator: central loop, two outer loops, narrow gaps ($g/2\pi$ = 15 mHz, V_m = 275 mm^3)
3 973 nm light through the crystal: optical pumping into $|\downarrow\rangle$ and absorption readout of the populations
4 microwave antennas at the outer loops: spin control pulses and transmission measurement
5 mixing-chamber plate of the dilution refrigerator (spin temperature 80 mK, n_{th} = 0.19)
6 $CaWO_4$ chip carrying the resonator
7 read-out volume under the inductor, where the microwave field is nearly uniform (coupling spread D)
8 niobium meander inductor (the part of the resonator that couples to the spins)
9 niobium capacitor pads
10 optical read-out beam through the read-out volume